Worksheet 4

Bivariate distributions and regression

Fully worked solutions

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1. Core applications

Master the essential definitions and calculations from the course.

Core application 15 min

Exercise 3 – Mayer line

For $(1,7),(2,10),(3,10),(4,14),(5,14)$, form two sub-clouds, calculate mean points A and B and determine the Mayer line.

Exercise 3 – Solution

$$A=\left(\frac{1+2+3}{3},\frac{7+10+10}{3}\right)=(2,9),$$

$$B=\left(\frac{4+5}{2},\frac{14+14}{2}\right)=(4.5,14).$$

$$a=\frac{14-9}{4.5-2}=2,\qquad b=5.$$

$$\boxed{y=2x+5}.$$

Auditable construction of the two centroids

Group	x	y	Mean x	Mean y
A	1, 2, 3	7, 10, 10	6/3=2	27/3=9
B	4, 5	14, 14	9/2=4.5	28/2=14

$$a=(14-9)/(4.5-2)=2,\quad b=9-2\times 2=5.$$

Checks: 2×2+5=9 and 2×4.5+5=14. The Mayer line is not necessarily the least-squares line.

2. Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice 35 min

Exercise 1 – Complete linear regression

Consider $(1,3),(2,5),(3,5),(4,8),(5,9)$.

1. Draw the scatter plot and describe direction, form and apparent strength.
2. Calculate $\bar{x}$, $\bar{y}$, $V(x)$, $V(y)$, $\mathrm{Cov}(x,y)$ and correlation r.
3. Estimate the least-squares line $\hat{y} = ax + b$.
4. Calculate fitted values and residuals $e_i = y_i - \hat{y}_i$.
5. Calculate SSE, SST and R^2 and interpret R^2.

Exercise 1 – Detailed solution

$$\bar{x} = 3, \qquad \bar{y} = 6, \qquad V(x) = 2, \qquad V(y) = 4.8.$$

$$\mathrm{Cov}(x,y) = 3, \qquad r = \frac{3}{\sqrt{2 \times 4.8}} \approx 0.96825.$$

$$a = \frac{3}{2} = 1.5, \qquad b = 6 - 1.5(3) = 1.5.$$

$$\boxed{\hat{y} = 1.5x + 1.5}.$$

x_i	1	2	3	4	5
y_i	3	5	5	8	9
$\hat{y}_i$	3	4.5	6	7.5	9
e_i	0	0.5	-1	0.5	0

$$SSE = 1.5, \qquad SST = 24,$$

$$R^2 = 1 - \frac{1.5}{24} = 0.9375.$$

About 93.75% of the observed variation in Y around its mean is accounted for by the fitted linear model.

Full computation table, including residuals and squared residuals:

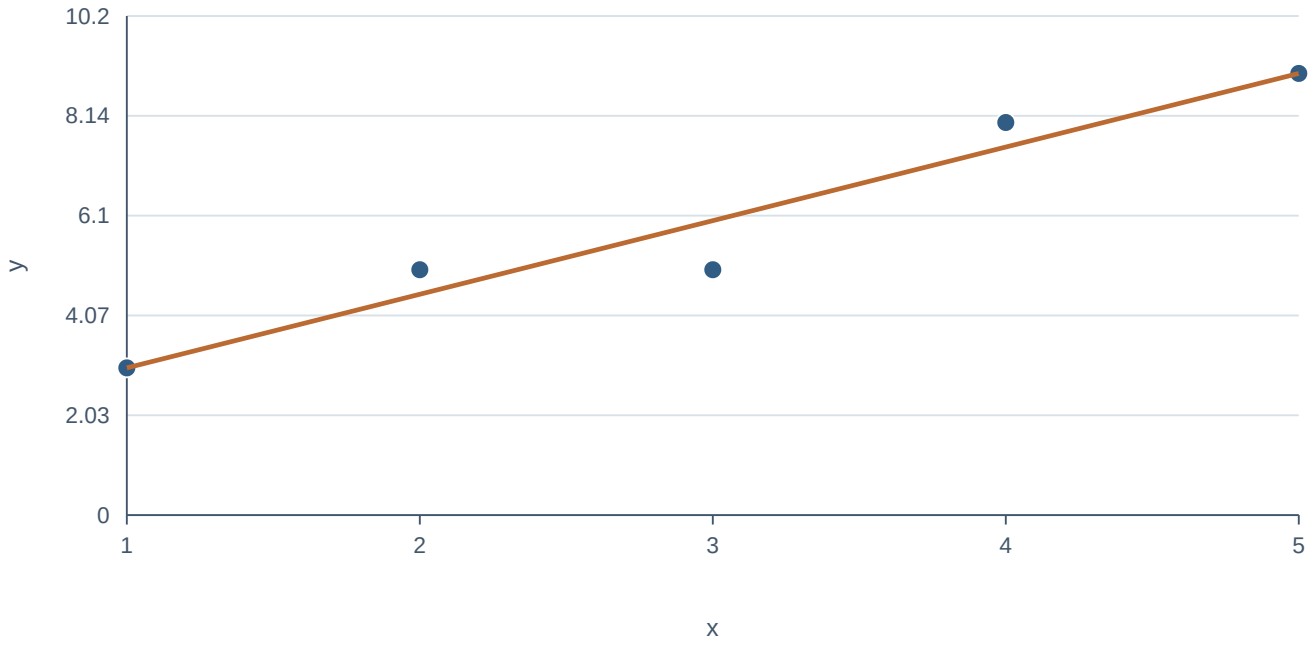
x_i	y_i	x_i^2	y_i^2	$x_i y_i$	$\hat{y}_i$	e_i	e_i^2
1	3	1	9	3	3	0	0
2	5	4	25	10	4.5	0.5	0.25
3	5	9	25	15	6	-1	1
4	8	16	64	32	7.5	0.5	0.25
5	9	25	81	45	9	0	0

x_i	y_i	x_i^2	y_i^2	$x_i y_i$	$\hat{y}_i$	e_i	e_i^2	
Σ	15	30	55	204	105	30	0	1.5

$$\bar{x} = 15/5 = 3, \quad \bar{y} = 30/5 = 6, \quad V(x) = 55/5 - 3^2 = 2, \quad V(y) = 204/5 - 6^2 = 4.8.$$

$$\mathrm{Cov}(x,y) = 105/5 - 3(6) = 3, \quad a = 3/2 = 1.5, \quad b = 6 - 1.5(3) = 1.5.$$

$$SSE = 0 + .25 + 1 + .25 + 0 = 1.5, \quad SST = 24, \quad R^2 = 1 - 1.5/24 = .9375.$$



Nuage de points / Scatter and OLS line

Assessment / exam practice 15 min

Exercise 4 – Two-stage forecast

Month	1	2	3	4	5	6	7	8
Advertising X	12	15	14	18	20	22	25	27
Sales Y	48	52	50	59	63	68	73	78

Given $\widehat{Y} = 2.064X + 21.899$ and $\widehat{X}_t = 2.155t + 9.429$, forecast month-10 advertising and sales, and identify where extrapolation occurs.

Exercise 4 – Solution

$$\widehat{X}_{10} = 2.155(10) + 9.429 \approx 30.98.$$

This exceeds the observed advertising maximum 27, so the first step extrapolates the time trend beyond month 8.

$$\widehat{Y}_{10} = 2.064(30.98) + 21.899 \approx 85.84.$$

The final sales forecast is about 85.84 thousand euros.

Complete two-stage calculation

t	1	2	3	4	5	6	7	8
X	12	15	14	18	20	22	25	27
Y	48	52	50	59	63	68	73	78

$$\widehat{X}_{10} = 2.155(10) + 9.429 = 21.55 + 9.429 = 30.979.$$

$$\widehat{Y}_{10} = 2.064(30.979) + 21.899 = 63.940656 + 21.899 = 85.839656.$$

Month 10 is outside the observed range 1–8, and predicted advertising 30.979 exceeds the observed maximum 27: this is a two-stage extrapolation requiring caution.

Assessment / exam practice 45 min

Exercise 5 — Two successive regressions — exam style

For six periods, observe period t, advertising x and sales y.

t	1	2	3	4	5	6
x	4	5	7	8	10	11
y	10	12	15	18	22	24

1. Build the complete t, x, y, t², x², y², tx, xy table and sums.
2. Compute means, variances, covariance, r and the regression of x on t.
3. Repeat for y on x.
4. Forecast x then y for t=8; discuss extrapolation.

Exercise 5 — Fully worked solution

t	x	y	t²	x²	y²	t×x	x×y
1	4	10	1	16	100	4	40
2	5	12	4	25	144	10	60
3	7	15	9	49	225	21	105
4	8	18	16	64	324	32	144
5	10	22	25	100	484	50	220
6	11	24	36	121	576	66	264
Σt=21	45	101	91	375	1853	183	833

Regression of x on t.

$$\bar{t} = 21/6 = 3.5,\ \bar{x} = 45/6 = 7.5,\ V(t) = 91/6 - 3.5^2 = 2.916667,\ V(x) = 375/6 - 7.5^2 = 6.25.$$

$$\mathrm{Cov}(t,x) = 183/6 - 3.5(7.5) = 4.25, \quad r_{tx} = \frac{4.25}{\sqrt{2.916667 \times 6.25}} = .995418.$$

$$a_1 = 4.25/2.916667 = 1.457143, \quad b_1 = 7.5 - a_1(3.5) = 2.4, \quad \widehat{x}_t = 1.457143t + 2.4.$$

Regression of y on x.

$$\bar{y} = 101/6 = 16.833333, \quad V(y) = 1853/6 - (101/6)^2 = 25.472222, \quad \mathrm{Cov}(x,y) = 833/6 - (45/6)(101/6) = 12.583333.$$

$$r_{xy} = \frac{12.583333}{\sqrt{6.25 \times 25.472222}} = .997292, \quad a_2 = 12.583333/6.25 = 2.013333, \quad b_2 = 16.833333 - a_2(7.5) = 1.733333.$$

$$\hat{y} = 2.013333x + 1.733333, \quad \widehat{x}_8 = 14.057143, \quad \hat{y}_8 = 2.013333(14.057143) + 1.733333 = 30.035048.$$

t=8 is beyond the observed 1–6 range. Both stages extrapolate; neither establishes causation.

3. Optional advanced practice

Explore additional interpretations and professional applications.

Advanced practice 20 min

Exercise 2 – Interpolation, extrapolation, causality and influence

Using Exercise 1, forecast Y at $X = 3.5$ and $X = 10$, classify each forecast, add point $(10, 30)$ and recalculate the regression line, then explain why high correlation does not establish causality.

Exercise 2 – Detailed solution

$$\hat{y}(3.5) = 1.5(3.5) + 1.5 = 6.75.$$

Because $3.5 \in [1, 5]$, this is interpolation.

$$\hat{y}(10) = 16.5.$$

Since 10 lies far outside the observed range, this is risky extrapolation.

After adding $(10, 30)$, the new line is approximately

$$\hat{y} = 3.04918x - 2.70492.$$

The slope rises from 1.5 to about 3.05, showing that the new observation is highly influential. High correlation remains an association and cannot by itself establish causality.

After adding (10,30), N=6, Σx=25, Σy=60, Σx²=155, Σy²=1104, Σxy=405.

$$V(x) = 155/6 - (25/6)^2 = 8.472222, \quad \mathrm{Cov}(x,y) = 405/6 - (25/6)(60/6) = 25.833333.$$

$$a = 25.833333/8.472222 = 3.049180, \quad b = 10 - 3.049180(25/6) = -2.704918.$$

The slope changes from 1.50 to 3.05: the additional point is influential.

Advanced practice 20 min

Extra 1 — Regression diagnosis and responsible decision

A model linking store traffic to temperature reports $r = 0.78$ and $R^2 = 0.61$. Its residual plot is U-shaped, one summer weekend has a very large residual, and the manager wants to forecast a temperature 12°C above the observed maximum.

1. Interpret r and R^2 correctly.
2. Diagnose the residual pattern.
3. Discuss the unusual weekend and the proposed forecast.
4. Recommend the next analytical steps.

Separate strength, functional form, unusual observations, extrapolation and causality.

Extra 1 — Regression diagnosis and responsible decision

Detailed solution

Correlation. $r = 0.78$ indicates a fairly strong positive linear association in the observed sample. It does not establish causality.

R^2. About 61% of the sample variation in traffic around its mean is explained by the fitted linear relationship. It does not mean that temperature “causes 61%” of traffic.

Residual diagnosis. A U-shape is systematic structure, indicating that a straight line misses curvature. Predictions may be biased at low and high temperatures.

Unusual weekend. Check data quality and contextual factors such as a festival, promotion or closure. Report sensitivity with and without the observation; do not delete it mechanically.

Extrapolation. Forecasting 12°C beyond the maximum is unsupported and particularly unsafe because curvature is already visible.

Next steps. Add a quadratic term or use a flexible curve, validate on held-out periods, include calendar and promotional controls, inspect influence statistics, and present prediction uncertainty.